Aditya Patil

Lead Software Engineer · Full-Stack · AI Workflow Automation

Pune, India | +91 78756 55996 | aadityapatil350@gmail.com | adityapatil.work | github.com/aadityapatil350

SUMMARY

Lead Software Engineer with 4+ years of experience building and scaling production systems for the renewable energy sector. Currently leading engineering initiatives at Renewalytics, where I architect, develop, deploy, and maintain mission-critical platforms serving 3,400+ MW of renewable assets across India.

I operate across the full technology stack — from product design and software architecture to cloud infrastructure, DevOps, and team leadership. My work includes forecasting and scheduling platforms, operational automation systems, DSM management, asset monitoring, and AI-driven workflows that help energy companies make faster and more accurate decisions. Passionate about building products from zero to scale, solving complex business problems, and leveraging AI to automate operational processes.

EXPERIENCE

Software Engineer → Lead Software Engineer Jun 2024 - Present

Renewalytics · Pune, India

- Promoted from Software Engineer to Lead Software Engineer within the first year based on shipped product impact.
- Own full-stack development and backend architecture for a renewable energy SaaS platform serving 3,400+ MW across India.
- Shipped **Reflux** within 3 months of joining, partnering with the ops team. Operational platform for solar and hydro forecasting revisions (CTU/STU to SLDC), scheduling, and reporting automation; now serves 40+ plants across 13+ clients.
- Shipped **Excel Flow** later the same year with one intern. Automated DGR reporting pipeline and ops dashboards for 33+ solar/wind/BESS plants; scrapes 30+ daily Excel sheets, generates next-day outputs, and surfaces live ops data.
- Took on **COPS** for Juniper Green Energy once Excel Flow workload eased; developed and deployed end-to-end as primary engineer, working directly with Juniper's Digital Transformation Leader. DSM penalties and asset management dashboard covering 14+ plants.
- Delivered an internal Invoice Management System for tracking and reconciling client invoices generated through Reflux.
- Containerized services with Docker, manage Linux infrastructure, and set up CI/CD with GitHub Actions.
- Integrated SCADA systems for realtime data exchange across multiple renewable assets.
- Currently building **RealSync CMS**, an enterprise realtime monitoring system with WebSockets, MQTT, OPC-UA ingestion, alarm systems, and AI-ready telemetry pipelines (in development).

Software Engineer Jul 2022 - Jan 2024

Climate Connect Digital

- Built ARS (Automatic Reporting System) for wind and solar plants; full-stack development in PHP.
- Owned end-to-end feature delivery from database design through UI polish.
- Maintained API integrations with third-party services and internal systems.
- Led frontend architecture decisions and component library work.

Full Stack Engineer Nov 2021 - Jul 2022

Climate Connect Digital

- Developed PHP-based features across the renewable energy product surface.
- Designed and implemented database schemas for new application modules.
- Shipped multiple production features independently.

Engineer Intern Jun 2021 - Nov 2021

Climate Connect Digital

- Onboarded into the engineering team with no prior professional software experience.
- Learned PHP and modern web development practices on the job.
- Contributed to production codebases within the first month.

SELECTED PROJECTS

Reflux — Operational platform for forecasting, scheduling, and reporting automation across 3,400+ MW. Next.js, PostgreSQL, MongoDB, Prisma, Docker.

Excel Flow — Automated DGR reporting pipeline and ops dashboard for 33+ solar/wind/BESS plants; 30+ daily Excel sheets processed via cron-based async workers. Node.js, PostgreSQL, Docker.

COPS DSM & CSM Portal — DSM penalties and asset management dashboard for Juniper Green Energy; developed and deployed end-to-end under their Digital Transformation Leader. Covers 14+ plants.

RealSync CMS (in development) — Realtime monitoring platform with SCADA, MQTT, OPC-UA, WebSockets, and alarm systems for renewable energy portfolios.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, PHP, SQL

Frontend: Next.js, React, Tailwind CSS, shadcn/ui

Backend: Node.js, Express, Prisma ORM, REST APIs

Databases: PostgreSQL, MongoDB, Redis

Infrastructure: Docker, Linux, Nginx, GitHub Actions, DigitalOcean, Azure

Protocols: MQTT, OPC-UA, WebSockets, REST

AI & Automation: AI workflow design, prompt engineering, Claude Code, agent-based automation

EDUCATION

PGDM, IT Management Final Semester

MIT School of Management, Pune

Diploma in Mechanical Engineering 2020